

APARNA SINGH

Software Developer | AI/ML Engineer | Data Science and Analytics

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SUMMARY

Software Developer and AI/ML Engineer with expertise in Python, Java, JavaScript, machine learning, deep learning, and cloud computing. B.Tech Information Technology student (CGPA 9.03) with hands-on experience in end-to-end ML pipelines, REST API development, LLM integration, and scalable web application development. Google GenAI Exchange Semi-Finalist and Top 10 finisher at RIFT '26 Hackathon. Proficient in TensorFlow, Scikit-learn, Keras, OpenCV, NLP, Computer Vision, Docker, CI/CD, and GCP.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), SQL, HTML5, CSS3, Bash

AI/ML and Deep Learning: TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face, LangChain, RAG, NLP, Computer Vision, LLM Integration, Classification, Clustering, Regression, Model Evaluation (Precision, Recall, F1, ROC-AUC)

Data Science and Analytics: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Dimensionality Reduction (PCA), Power BI, Excel

Cloud and DevOps: Google Cloud Platform (GCP), Vertex AI, BigQuery, Cloud Run, Docker, Git, GitHub Actions, CI/CD Pipeline

Web and Backend Development: Next.js, Node.js, REST APIs, Firebase, Gemini API, Google Maps API, Google Translation API

Databases: MySQL, Firebase Firestore, BigQuery, SQL

Core Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

Bharati Vidyapeeth College of Engineering, Pune | July 2023 – Present

Bachelor of Technology, Information Technology | CGPA: 9.03 / 10.0

Tender Hearts School, Lucknow | February 2022

Higher Secondary Certificate (Class XII)

EXPERIENCE

AI/ML Developer — Google GenAI Exchange Hackathon 2025 | Semi-Finalist | 2025

Tech Stack: Next.js, Node.js, Python, Firebase, Gemini API, Vertex AI, BigQuery, RAG, Google Cloud Platform

- Built KarigarSetu, an AI-powered full-stack cultural marketplace using Next.js, Node.js, and the Gemini API with Retrieval-Augmented Generation (RAG) and LRU caching for low-latency conversational search, reducing average query response time by 40%.
- Architected a scalable GCP backend using Vertex AI for ML model inference and BigQuery for large-scale analytics; integrated Google Translation and Maps APIs supporting 5+ regional languages and location-aware discovery for artisans across India.

Software Developer — Smart India Hackathon (SIH) 2025 | RIFT '26 Hackathon — Top 10 | 2025 – 2026

National-level hackathons: agriculture domain (SIH) | health-tech application (RIFT '26, 24-hour open innovation sprint)

- Designed system architecture and delivered a functional prototype for an agriculture-sector problem statement at Smart India Hackathon 2025, a national government software engineering competition with 50,000+ registered teams.
- Shipped a health-tracking web application at RIFT '26 Hackathon (PW IOI), placing in the Top 10 out of 50+ teams within a 24-hour sprint.

PROJECTS

CreditWise — Loan Approval Prediction System | Python, Machine Learning, GitHub

Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook | Supervised Learning

- Constructed an end-to-end supervised ML pipeline for binary loan-approval classification using three algorithms — KNN, Logistic Regression, and Naive Bayes — with full EDA, feature engineering, and data preprocessing on real-world financial data.
- Achieved 92% classification accuracy with Logistic Regression; assessed all three models using Precision, Recall, F1-Score, and ROC-AUC, producing results suitable for automated financial decision-support systems.

Text Summarizer — AI-Powered NLP Summarization Tool | Python, NLP, GitHub

Python, PyTorch, Hugging Face Transformers, Flask | NLP, Transformer Models

- Created an AI-powered Text Summarizer using NLP and Transformer models to generate concise, accurate summaries from lengthy documents.
- Integrated Hugging Face Transformers with a Flask web interface, enabling efficient abstractive summarization via a lightweight REST API endpoint.

CERTIFICATIONS

- Cloud Computing — IIT Madras, NPTEL/Swayam (November 2024)
- Database Management Systems (DBMS) — IIT Kharagpur, NPTEL/Swayam (March 2025)
- GenAI Powered Data Analytics Job Simulation — Tata Forage (November 2025)

ACTIVITIES AND LEADERSHIP

- Technical Member, GeeksforGeeks Campus Body, Bharati Vidyapeeth — organized coding contests, DSA workshops, and tech talks (April 2025 – Present).
- UNESCO Innovation Program 2025 — Contributed to the Moody Project, supporting problem definition, solution design, and social-impact planning.